

KCPC-CS150

Problem Sheet

Data Structures I – Disjoint Set Union (DSU)

Developed by the KCPC Internal Committee, this problem sheet accompanies the *Data Structures I* workshop. Problems are divided into three categories:

- Warm-up problems (basic graph connectivity)
 - Core practice set (standard DSU and MST applications)
 - Extended homework (advanced graph modelling and DSU techniques)
-

I. Warm-up Problems

These problems introduce the concept of graph connectivity and connected components.

1. **AtCoder Practice2 A — Disjoint Set Union**
(Classic DSU implementation)

https://atcoder.jp/contests/practice2/tasks/practice2_a

II. Core Workshop Practice Set

These problems should be attempted during the workshop.

They focus on using **Disjoint Set Union (Union-Find)** and understanding graph connectivity and minimum spanning trees.

1. **CSES — Road Construction**
(Dynamic connectivity with DSU)
<https://cses.fi/problemset/task/1676/>
 2. **CSES — Road Reparation**
(Minimum Spanning Tree — Kruskal)
<https://cses.fi/problemset/task/1675/>
 3. **LeetCode 684 — Redundant Connection**
(Cycle detection with DSU)
<https://leetcode.com/problems/redundant-connection/>
 4. **Codeforces 977E — Cyclic Components**
(Detecting special connected components)
<https://codeforces.com/problemset/problem/977/E>
-

III. Extended Homework

Important. The following problems are significantly more difficult. They should only be attempted after completing all problems in the warm-up and core sections.

These exercises introduce more advanced graph modelling and DSU-based techniques.

Matthew's Note. If you have any questions about the problems or underlying concepts, feel free to contact me or the Director of Competitive Programmes (Marcell).

1. **CSES — Building Roads**
(Connected components and graph construction)
<https://cses.fi/problemset/task/1666/>
2. **Codeforces 1151B — Dima and a Bad XOR**
(Graph reasoning + constructive logic)
<https://codeforces.com/problemset/problem/1151/B>
3. **Codeforces 1514D — Cut and Stick**
(Advanced data structures + query processing)
<https://codeforces.com/problemset/problem/1514/D>
4. **Codeforces 343D — Water Tree**
(Tree operations + advanced data structures)
<https://codeforces.com/problemset/problem/343/D>
5. **Codeforces 500B — New Year Permutation**
(Graph connectivity + permutation reconstruction)
<https://codeforces.com/problemset/problem/500/B>
6. **HackerRank — Merging Communities**
(Classic DSU with component size queries)
<https://www.hackerrank.com/challenges/merging-communities>

Guidance:

- Learn the Disjoint Set Union structure (Union-Find).
- Use **path compression** and **union by rank / size**.
- Recognise when connectivity queries can be solved with DSU instead of graph traversal.
- For MST problems, apply **Kruskal's algorithm**.